LABORATORY OBSERVATION / RECORD

Course:	Full Stack Web Development	**Experiment No.:**	TASK 3
Course Code:	23CM4121	**Date:**	30-08-2026
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1. TITLE

Interactive To-Do List Application Using JavaScript, DOM Manipulation, and Event Listeners

2. AIM

To develop an interactive To-Do List web application using JavaScript, DOM manipulation, and event listeners, which allows the user to add, complete, and delete tasks dynamically.

3. OBJECTIVE

- To provide a text box and an Add Task button for entering new tasks.
- To dynamically create a new task item and add it to a list when the Add Task button is clicked.
- To provide a Complete button for each task that visually marks the task as completed when clicked.
- To provide a Delete button for each task that removes the corresponding task from the webpage when clicked.
- To display an appropriate message when the task list is empty.
- To demonstrate DOM element selection, dynamic element creation, dynamic modification of the DOM, and event handling.

4. THEORY

DOM Element Selection: document.getElementById() is used to select existing elements such as the input box, Add Task button, task list, and the empty-state message, so that JavaScript can read their values or modify them.

Dynamic Element Creation: document.createElement() is used to create new elements such as the list item (li), the task text (span), and the Complete/Delete buttons entirely in JavaScript, at the moment a new task is added.

Dynamic Modification of the DOM: Properties such as className and textContent are used to set the CSS class and visible text of newly created elements, and classList.toggle() is used to add or remove the "completed" style on a task's text.

Event Listeners and Click Handling: addEventListener("click", ...) attaches functions to the Add Task, Complete, and Delete buttons so that they respond when clicked. A "keydown" listener on the input box also allows a task to be added by pressing Enter.

Adding and Removing Elements: appendChild() inserts newly created elements (the task item, its text, and its buttons) into the task list, while element.remove() deletes a task item from the DOM when its Delete button is clicked.

Interactive Webpage Development: Combining the above techniques allows the page to respond immediately to user actions without reloading, updating the empty-list message automatically as tasks are

added or removed.

5. ALGORITHM / PROCEDURE

1. Create the HTML page with a text input (id="taskInput"), an Add Task button (id="addBtn"), an empty-state message (id="emptyMessage"), and an empty unordered list (id="taskList").
2. Link an external CSS file to style the container, input box, buttons, task items, and the completed-task text.
3. In the JavaScript file, select the input box, Add Task button, task list, and empty message using document.getElementById().
4. Define an updateEmptyState() function that shows the empty message when the task list has no children, and hides it otherwise.
5. Define an addTask() function that reads and trims the value from the input box, and shows an alert if it is empty.
6. Inside addTask(), dynamically create a list item, a span for the task text, a button container, a Complete button, and a Delete button using document.createElement().
7. Attach a click event listener to the Complete button that toggles a "completed" CSS class on the task text.
8. Attach a click event listener to the Delete button that removes the task item from the DOM and calls updateEmptyState().
9. Append the buttons to the button container, and append the text and button container to the list item, then append the list item to the task list.
10. Clear the input box, refocus it, and call updateEmptyState() after a task is added.
11. Attach a click event listener to the Add Task button that calls addTask(), and a keydown listener on the input box that calls addTask() when the Enter key is pressed.
12. Save the HTML, CSS, and JavaScript files and open the HTML file in a browser to verify adding, completing, and deleting tasks, as well as the empty-list message.

6. PROGRAM

index.html

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>To-Do List Application</title>
    <link rel="stylesheet" href="style3.css">
</head>
<body>

    <div class="container">
        <h1>My To-Do List</h1>

        <div class="input-group">
            <input type="text" id="taskInput" placeholder="Enter a task...">
            <button id="addBtn">Add Task</button>
        </div>

        <p id="emptyMessage">No tasks available.</p>
        <ul id="taskList"></ul>
    </div>

    <script src="script3.js"></script>
</body>
</html>
```

style3.css

```
body {
    font-family: Arial, sans-serif;
    background-color: #f4f6f8;
    padding: 30px;
    display: flex;
    justify-content: center;
}

.container {
    background-color: #ffffff;
    width: 480px;
    padding: 25px;
    border-radius: 8px;
    box-shadow: 0 4px 12px rgba(0, 0, 0, 0.1);
}

h1 {
    text-align: center;
    color: #333333;
    margin-top: 0;
}

.input-group {
    display: flex;
    gap: 10px;
    margin-bottom: 20px;
}

#taskInput {
    flex: 1;
    padding: 10px 12px;
    font-size: 14px;
    border: 1px solid #ccc;
    border-radius: 4px;
    outline: none;
}

#taskInput:focus {
    border-color: #007bff;
}

#addBtn {
    padding: 10px 16px;
    background-color: #007bff;
    color: white;
    border: none;
    border-radius: 4px;
```

```
    cursor: pointer;
    font-weight: bold;
}

#addBtn:hover {
    background-color: #0056b3;
}

#taskList {
    list-style: none;
    padding: 0;
    margin: 0;
}

.task-item {
    display: flex;
    align-items: center;
    justify-content: space-between;
    background-color: #fafafa;
    padding: 10px 12px;
    margin-bottom: 8px;
    border: 1px solid #e0e0e0;
    border-radius: 4px;
}

.task-text {
    flex: 1;
    font-size: 15px;
    color: #333;
    word-break: break-word;
    margin-right: 10px;
}

/* Completed task style */
.completed {
    text-decoration: line-through;
    color: #888888;
}

.btn-group {
    display: flex;
    gap: 6px;
}

.complete-btn {
    background-color: #28a745;
    color: white;
    border: none;
    padding: 6px 12px;
    border-radius: 4px;
    cursor: pointer;
}

.complete-btn:hover {
    background-color: #218838;
}

.delete-btn {
    background-color: #dc3545;
    color: white;
    border: none;
    padding: 6px 12px;
    border-radius: 4px;
    cursor: pointer;
}

.delete-btn:hover {
    background-color: #c82333;
}

#emptyMessage {
    text-align: center;
    color: #777777;
    font-style: italic;
    margin: 15px 0;
}
```

script3.js

```
// 1. DOM Element Selection
const taskInput = document.getElementById("taskInput");
const addBtn = document.getElementById("addBtn");
const taskList = document.getElementById("taskList");
const emptyMessage = document.getElementById("emptyMessage");

// Helper function to toggle empty state message
function updateEmptyState() {
    if (taskList.children.length === 0) {
        emptyMessage.style.display = "block";
    } else {
        emptyMessage.style.display = "none";
    }
}

// Function to create and append a new task item
function addTask() {
    const taskText = taskInput.value.trim();

    if (taskText === "") {
        alert("Please enter a task before adding.");
        return;
    }

    // 2. Dynamic Element Creation
    const li = document.createElement("li");
    li.className = "task-item";

    const textSpan = document.createElement("span");
    textSpan.className = "task-text";
    textSpan.textContent = taskText;

    const btnContainer = document.createElement("div");
    btnContainer.className = "btn-group";

    const completeBtn = document.createElement("button");
    completeBtn.className = "complete-btn";
    completeBtn.textContent = "Complete";

    const deleteBtn = document.createElement("button");
    deleteBtn.className = "delete-btn";
    deleteBtn.textContent = "Delete";

    // 3. Event Listeners for Dynamic Elements
    // Toggle task completion
    completeBtn.addEventListener("click", function () {
        textSpan.classList.toggle("completed");
    });

    // Delete task item
    deleteBtn.addEventListener("click", function () {
        li.remove();
        updateEmptyState();
    });

    // 4. Construct the DOM tree
    btnContainer.appendChild(completeBtn);
    btnContainer.appendChild(deleteBtn);

    li.appendChild(textSpan);
    li.appendChild(btnContainer);

    taskList.appendChild(li);

    // Clear input and update empty list message
    taskInput.value = "";
    taskInput.focus();
    updateEmptyState();
}

// 5. Event Listeners
addBtn.addEventListener("click", addTask);

// Allow adding task by pressing the 'Enter' key
taskInput.addEventListener("keydown", function (event) {
    if (event.key === "Enter") {
        addTask();
    }
```

```
});
```

7. CODE EXPLANATION

Code Element	Description
document.getElementById(...)	Selects the input box, Add Task button, task list, and empty-state message from the DOM.
updateEmptyState()	Checks taskList.children.length and shows or hides the "No tasks available." message accordingly.
taskInput.value.trim()	Reads the text entered by the user and removes leading/trailing whitespace before validating it.
document.createElement("li"/"span"/"div"/"button")	Dynamically creates the task item, its text, its button container, and its Complete/Delete buttons.
li.className / textSpan.className	Assigns CSS classes (task-item, task-text) to newly created elements for styling.
completeBtn.addEventListener("click", ...)	Registers a click handler that toggles the "completed" class on the task text, giving it a strikethrough style.
textSpan.classList.toggle("completed")	Adds the completed class if absent, or removes it if already present, switching the task's completed state.
deleteBtn.addEventListener("click", ...)	Registers a click handler that removes the task item from the page and refreshes the empty-state message.
li.remove()	Removes the corresponding list item element from the DOM.
appendChild(...)	Builds the task item by nesting the text and buttons inside it, then adds the finished item to the task list.
addBtn.addEventListener("click", addTask)	Runs addTask() whenever the Add Task button is clicked.
taskInput.addEventListener("keydown", ...)	Runs addTask() when the Enter key is pressed while typing in the input box.

8. TEST CASES AND EXECUTION DATA

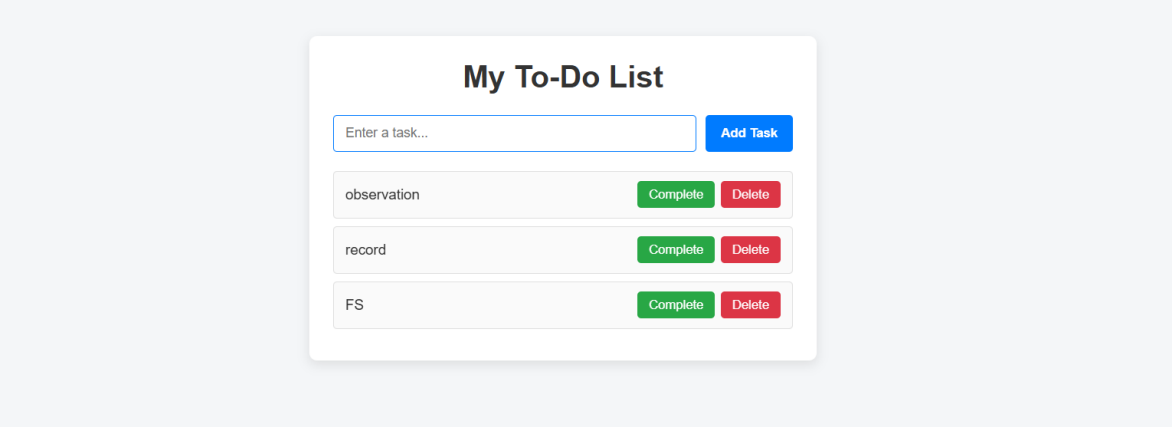
The application was verified in a web browser by adding, completing, and deleting tasks, and checking the displayed output against the expected behaviour.

Test Case	Action	Expected Result	Actual Result	Status
TC-01	Page loaded with no tasks added	"No tasks available." message is shown	Empty message displayed correctly	Pass
TC-02	Click Add Task with the input box empty	An alert asks the user to enter a task; no task is added	Alert shown; no task added	Pass
TC-03	Enter "observation" and click Add Task	A new task item "observation" appears in the list	Task added and displayed correctly	Pass
TC-04	Add "record" and "FS" as additional tasks	All three tasks are listed in the order added	All tasks listed correctly	Pass
TC-05	Press Enter after typing a task instead of clicking Add Task	Task is added the same way as clicking the button	Task added correctly via Enter key	Pass
TC-06	Click Complete on a task	Task text shows a strikethrough style	Strikethrough style applied	Pass
TC-07	Click Complete again on the same task	Strikethrough style is removed (toggled back)	Style removed correctly	Pass
TC-08	Click Delete on a task	The task is removed from the list	Task removed correctly	Pass
TC-09	Delete all remaining tasks	"No tasks available." message reappears	Empty message reappeared correctly	Pass

9. OUTPUT

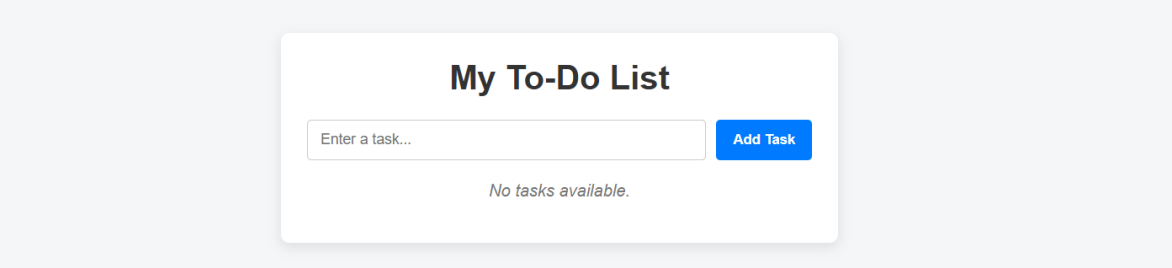
Output 1: Task List with Multiple Tasks

Displays the To-Do List after adding three tasks (“observation”, “record”, and “FS”), each with its own Complete and Delete buttons.



Output 2: Empty Task List

Displays the To-Do List in its initial state, with no tasks added, showing the “No tasks available.” message below the input box.



10. RESULT

Thus, an interactive **To-Do List application** was successfully developed using JavaScript, DOM manipulation, and event listeners. Tasks can be added dynamically, marked as completed with a visual strikethrough style, and removed from the list, with an appropriate message shown when the list is empty. The application was verified in a web browser and all features functioned as expected.